What is LoRA (lov

Machine learning

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By Joshua Noble

This article was featured in the Think newslette

What is LoRA?

Low-rank adaptation (LoRA) is a tech learning models to new contexts. It c uses by adding lightweight pieces to changing the entire model. A data sc that a model can be used rather than new model.

The technique was originally published by Edv their paper LoRA: Low-Rank Adaptation Of Lar showed that models reduced and retrained us variety of benchmark tasks. The model perfor full fine-tuning and by using a significantly sm

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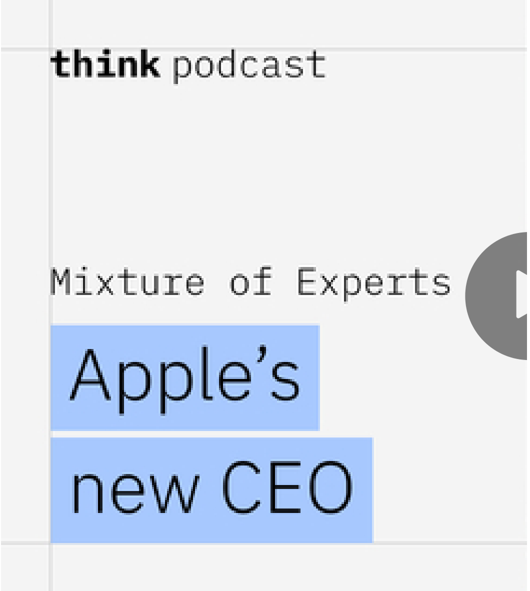

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What does LoRA do?

Large and complex machine learning models,
(LLMs) like ChatGPT, take a long time and num
trillions of parameters that are set to specific
model might be powerful and accurate in gene
carry out specific tasks.

Getting a model to work in specific contexts ca
all its parameters. With the number of parame
expensive and time-consuming. LoRA provide
retraining it.

As an example, a full fine-tuning of the GPT-3
parameters because of the size of its training
for GPT-3 can be reduced to roughly 18 millio
requirements by roughly two thirds.

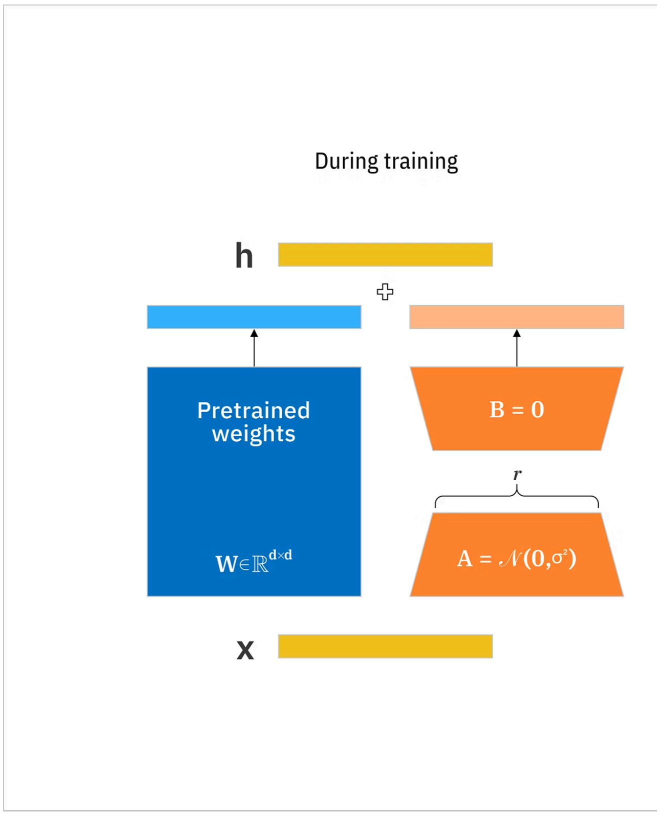
LoRA is not the only efficient fine-tuning meth
(QLoRA), a fine tuning technique that combine
a low-precision storage method. This helps ke
sure the model is still highly performant and a

How LoRA works

Instead of retraining the whole model, LoRA fr
the model as they are. Then, on top of this orig
called a low-rank matrix, which is then applied
context. The low-rank matrix adjusts for the w
match the desired use case.

LoRA leverages the concept of lower-rank mat
extremely efficient and fast. Traditionally fine-
model. LoRA focuses on modifying a smaller s
reduce computational and memory overhead.

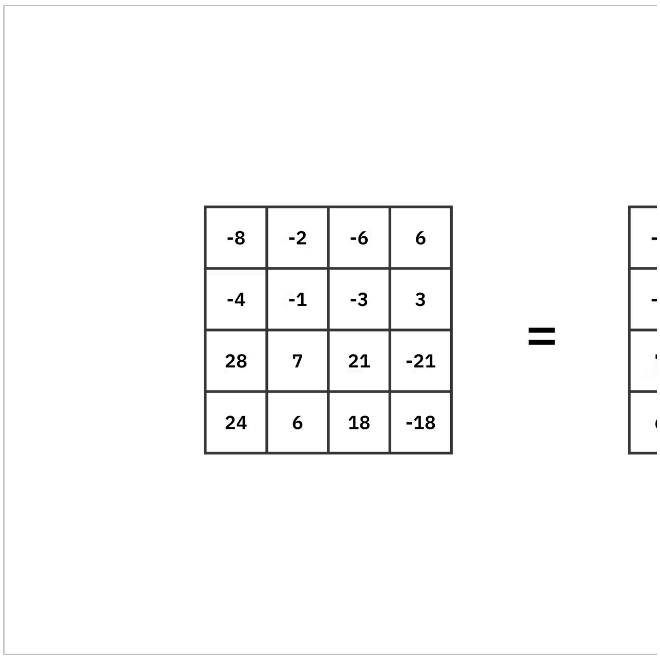
The diagram shows how LoRA updates the ma
pretrained weights by using smaller matrices
smaller weights are merged into a new weight
original weights of the pretrained model.



LoRA is built on the understanding that large r
structure. By leveraging smaller matrices, whi
these models effectively. This method focuses
changes can be represented with fewer param
more efficient.

Matrices are an important part of how machin
Low-rank matrices are smaller and have many
matrices. They do not take up much memory a
together and this makes them faster for comp

A high-rank matrix can be decomposed into tw
decomposed into a 4 x 1 and a 1 x 4 matrix.



LoRA adds low-rank matrices to the frozen ori
matrices are updated through gradient descen
weights of the base model. These matrices co
when generating results. The multiplied chang
to get the final fine-tuned model. This process
with minimal computing power and training tin

In essence, LoRA keeps the original model und
each layer of the model. This significantly redu
and the GPU memory requirement for the train
challenge when it comes to fine-tuning or train

To implement LoRA fine tuning with HuggingF
can use the parameter-efficient fine-tuning (P
the model and use them as the update matric
HuggingFace or GitHub. This library provides v
model. Some of the most commonly used par

r: the rank of the update matrices, expressed
smaller update matrices with fewer trainable

target_modules: The modules (for example, a
matrices.

lora_alpha: LoRA scaling factor.

Advantages of LoRA

One of the key advantages of LoRA is that a ba
many small LoRA modules for new tasks. The
switch tasks by replacing the LoRA weight ma
models, a single model can be used in differer
gains of fine-tuning.

LoRA makes training more efficient and lower
do not need to calculate the gradients or main
parameters. Instead, the process requires opt
matrices.

The linear design of LoRA allows data scientis
frozen pretrained model weights when deploy
compared to a fully fine-tuned model by const

LoRA can be combined with other techniques
prefix-tuning, making it flexible.

Tradeoffs

While LoRA provides a significant reduction in
a tradeoff as well. The process creates inform
Because LoRA reduces the fullweight matrix in
lost in the process. This is comparable to mod

loss is often minimal because deep learning m
meaning that they contain more parameters tl
parameterized" means that these models are
training data. Not all the parameters strictly m
resiliency within the parameters. Removing th
so the rank of update matrices can be tuned a

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Footnotes

1. Hu, Edward, et al, LoRA: Low-Rank Adaptation of Large Language Models, 2021